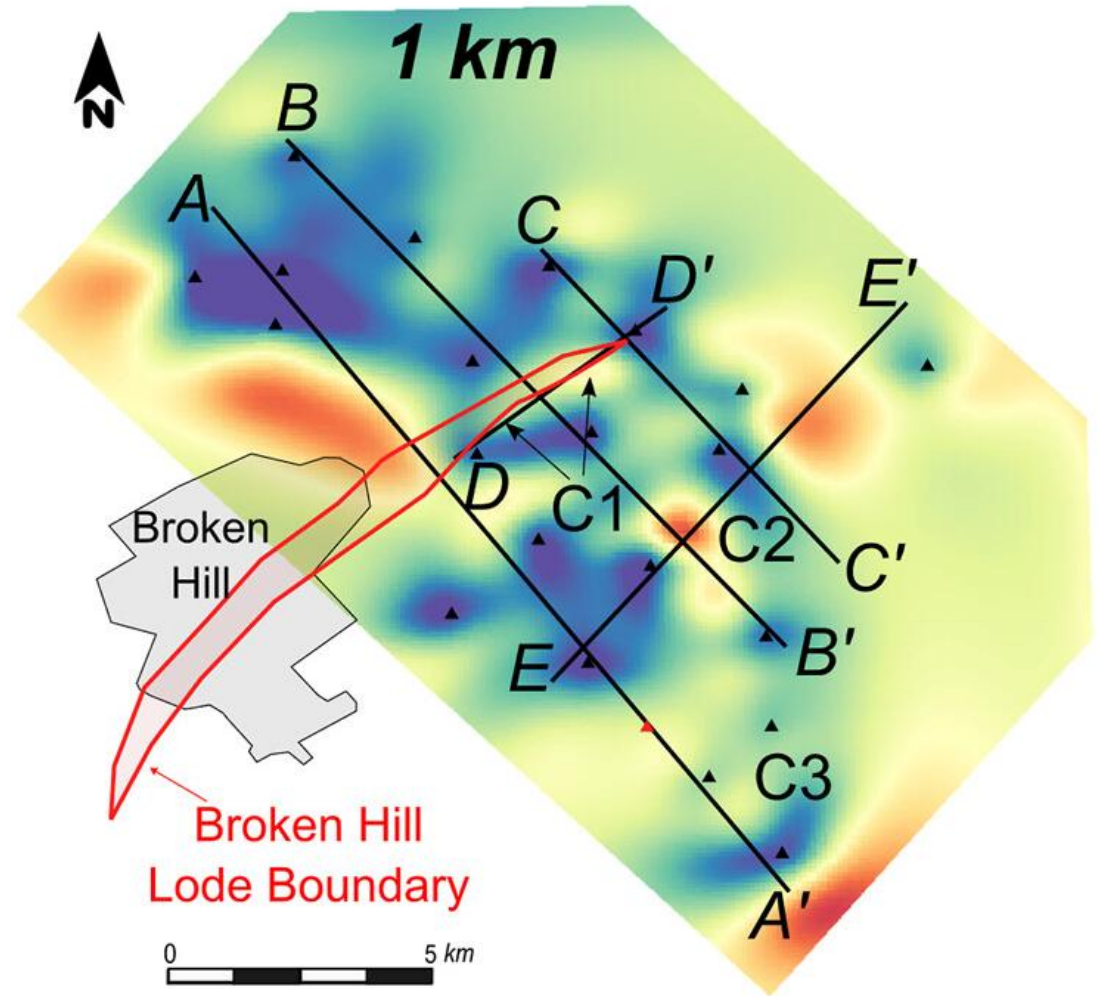


Broken Hill - ABIC

Regional SSQ analysis, forward validation, mesh and model-norm tests, galvanic distortion, model appraisal, and published ModEM results

Because ABIC determines the regularization parameter and final model statistically and showed no dependence on the tested initial models, the inversion tests focused on mesh discretization and galvanic-distortion correction at Broken Hill.

20 September 2026



Contents

1	Regional SSQ analysis	4
2	Forward validation and linearized sensitivity analysis	5-6
3	H0, H1, H2, T0, Difference-L1, and distortion	7-12
4	Published ModEM comparison	13
5	Model appraisal	14-18
6	Conclusion and appendix	19-23

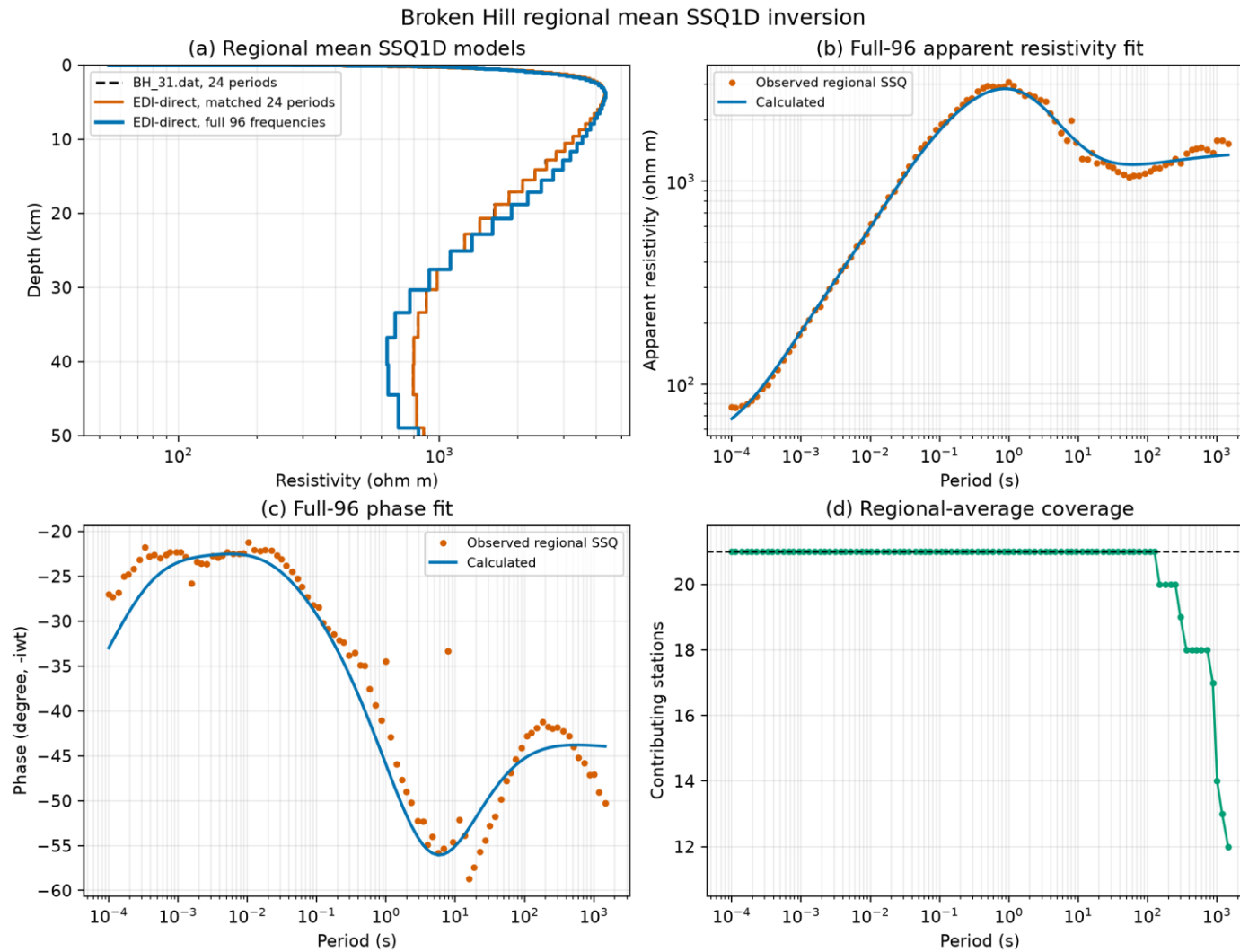
Inversion tests including Difference-L1

Summary

Case	Mesh	Iteration	RMS
H0	Uniform brick DHEXA	17	1.7446
H1	Non-conforming DHEXA	11	1.7771
H2	Nested non-conforming DHEXA	13	1.7374
T0	Tetrahedral	19	1.69145
H2-L1	Nested non-conforming DHEXA	16	1.8026

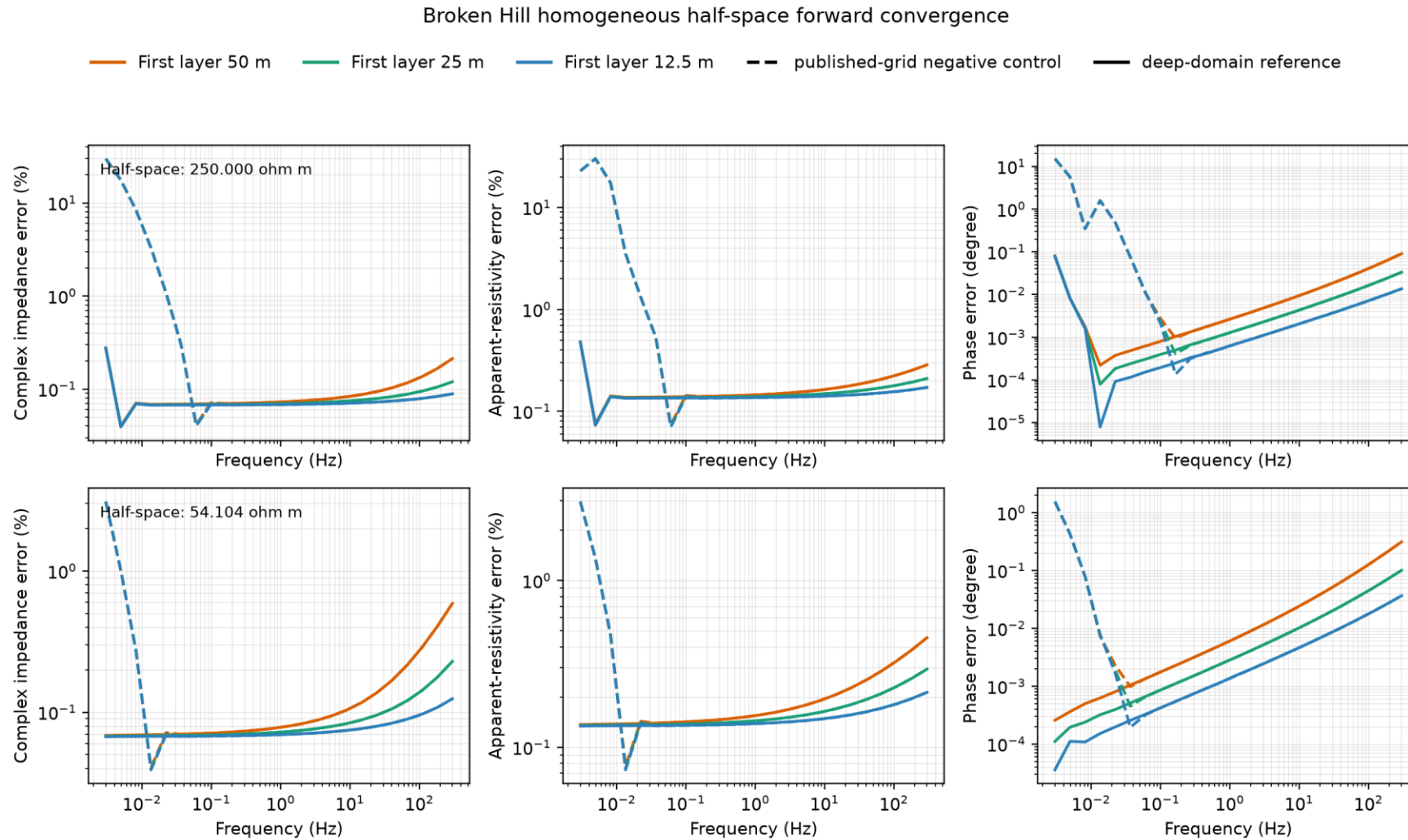
All tests use the FEMTIC forward solver and the ABIC inversion method.

Regional mean SSQ1D inversion



The regional 1-D background from 96-frequency EDI data increases from 54 Ω m at the surface to about 4.3 k Ω m near 4 km depth, then decreases to about 3.7 k Ω m at 10 km.

Homogeneous half-space forward convergence



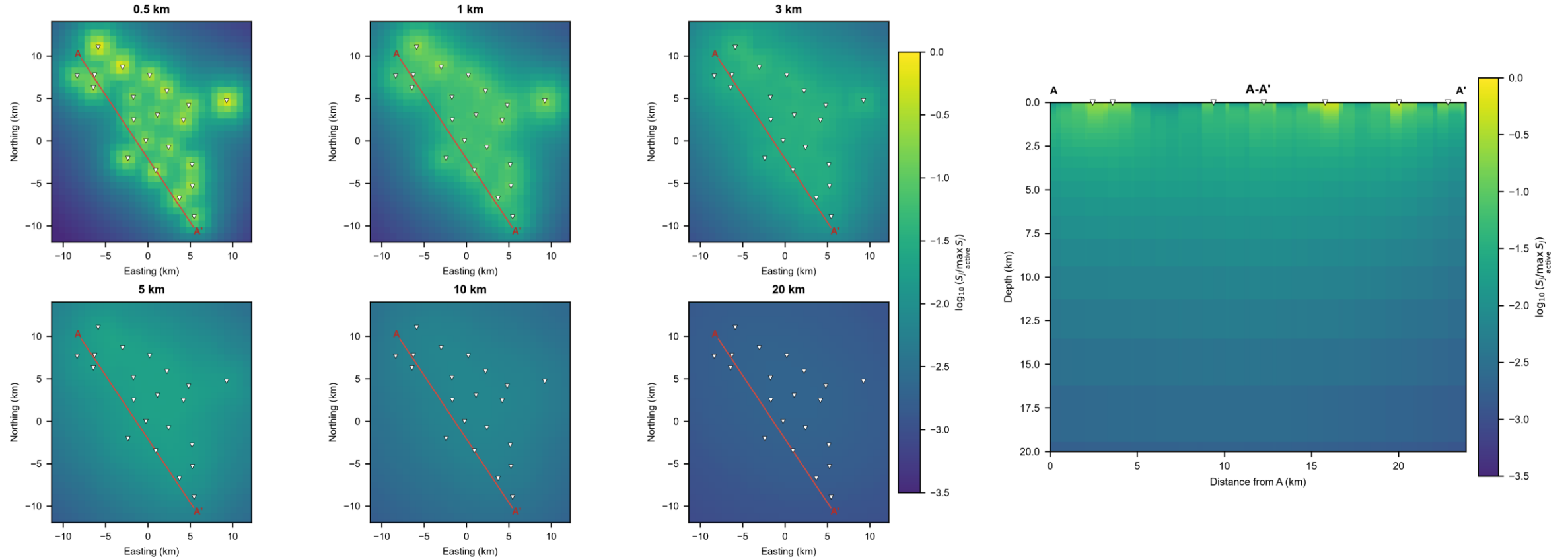
For 12.5, 25, and 50 m first-layer meshes with a 500 km bottom, the off-diagonal complex-response RMS error is 0.078-0.184%; the adopted 25 m mesh gives 0.099% and 0.093% for the 54.1 and 250 Ω m tests.

Uniform-grid homogeneous half-space sensitivity

Forward validation and linearized sensitivity analysis

$$S_j = \sum_i |J_{ij}|; \text{ displayed value} = \log_{10}[S_j / \max_{\text{active}}(S_j)]$$

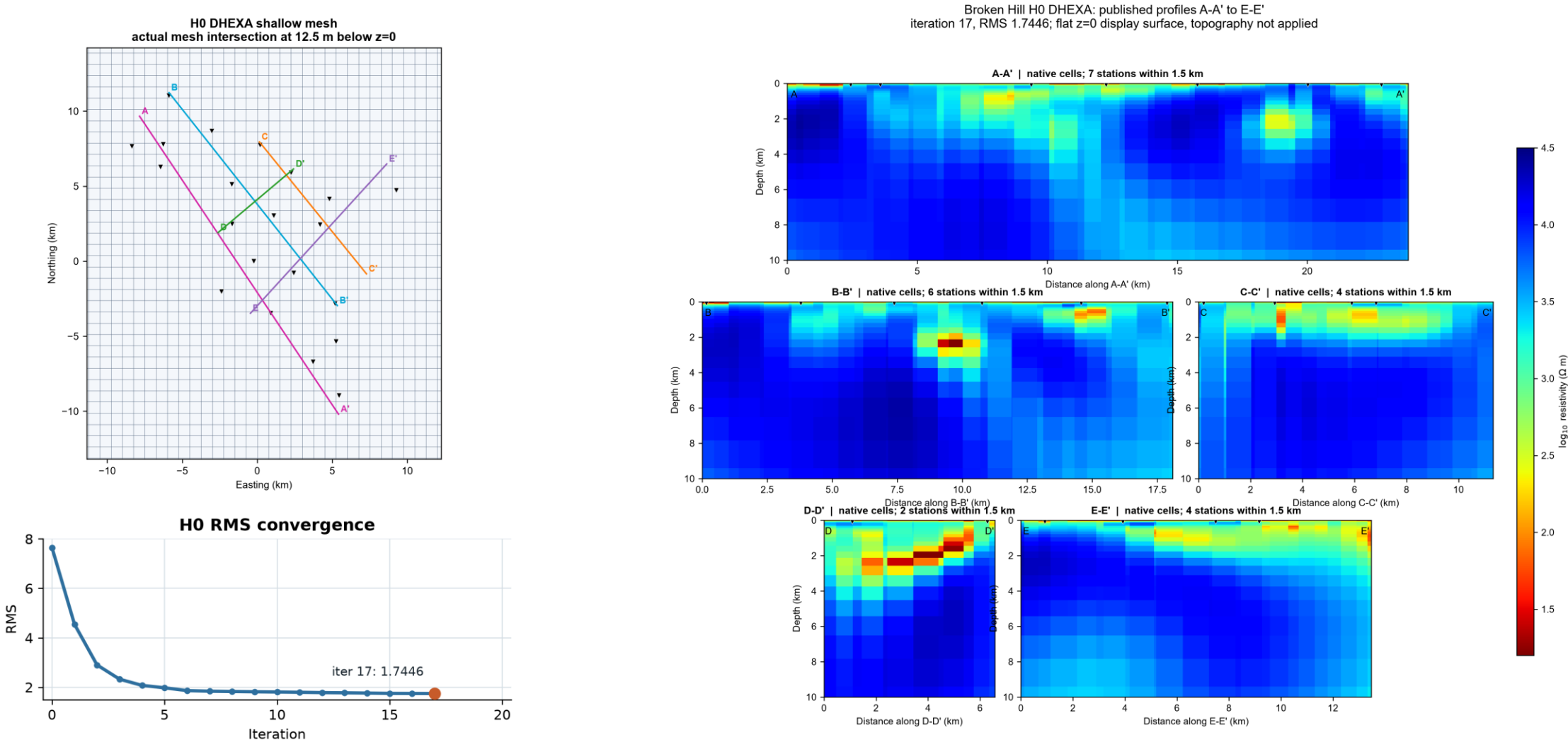
H0 uniform-brick DHEXA · 250 Ω m half-space



As expected, mean block-integrated sensitivity decreases with depth; relative to 0.5 km, it is 47.3% at 3 km, 30.9% at 5 km, and 13.8% at 10 km.

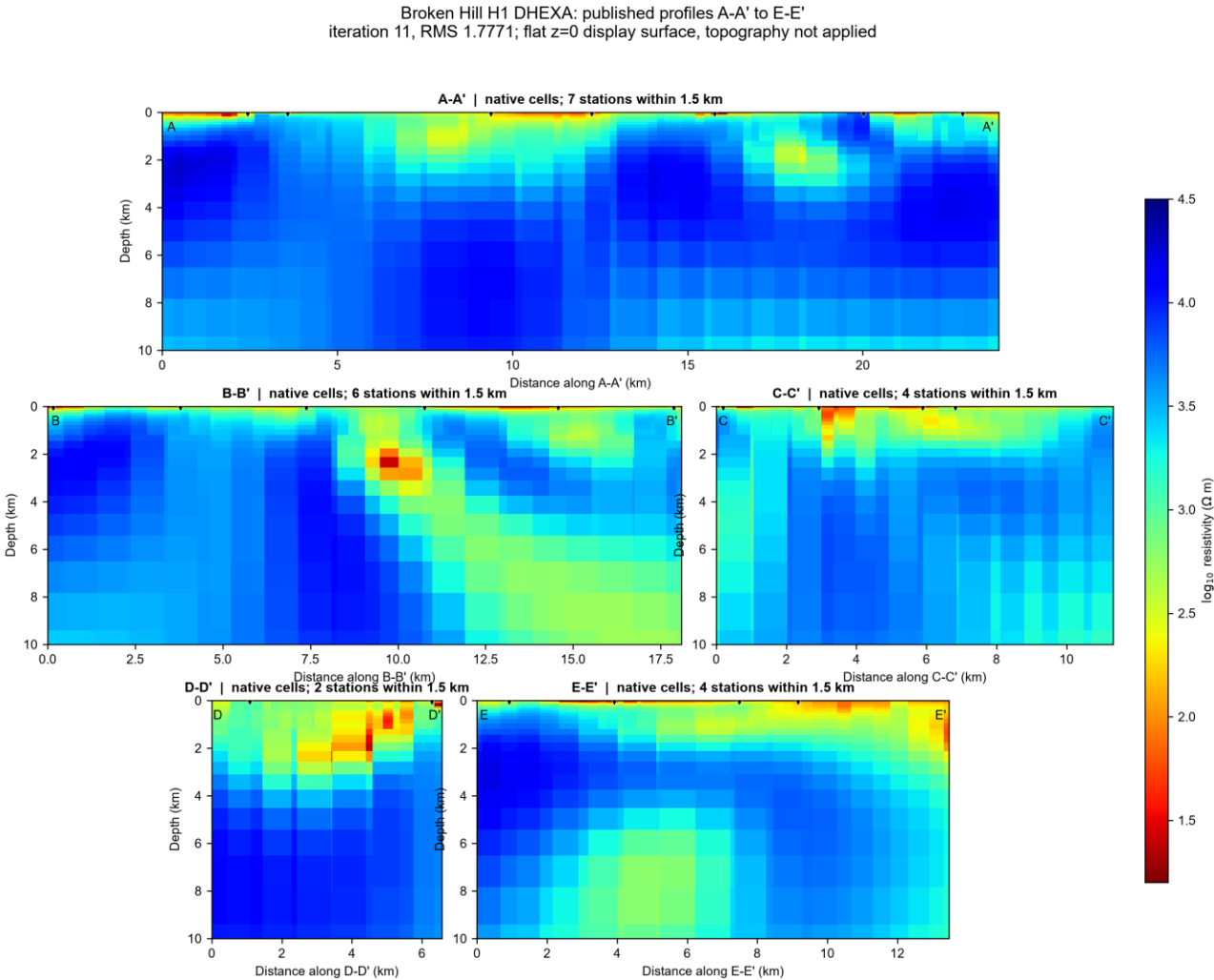
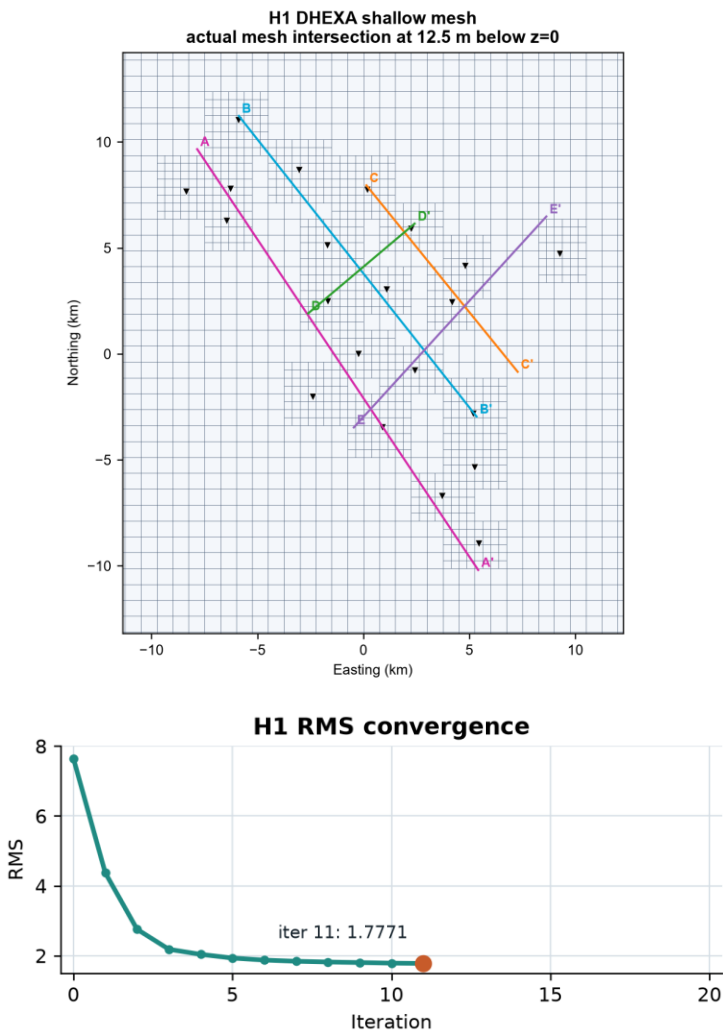
Although no stations lie southwest of A–A', sensitivity within 3–5 km on either side at 1–5 km depth remains comparable to the central high-sensitivity area at the same depths.

H0 uniform brick DHEXA



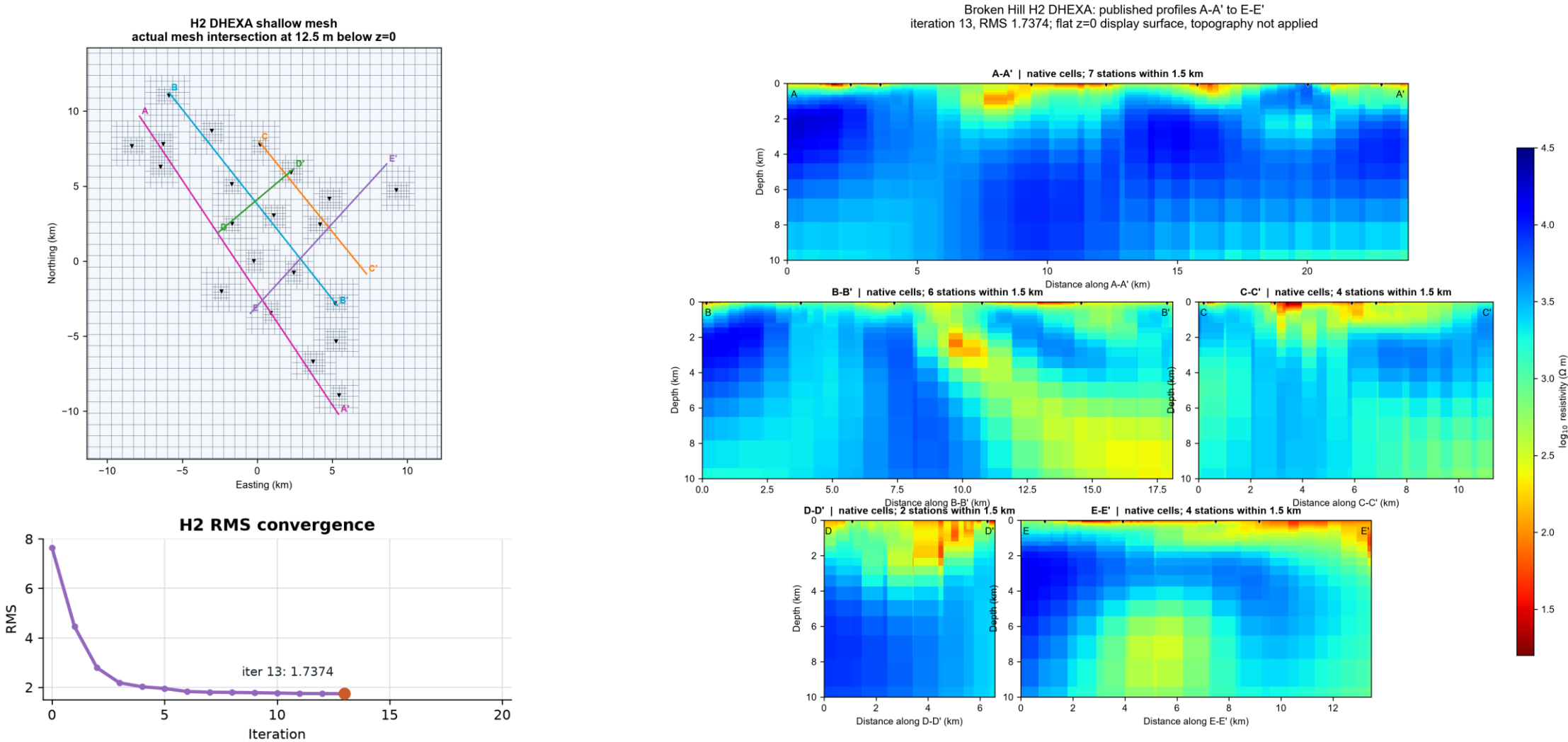
H0 uses 750 m surface cells and converged at iteration 17 with RMS 1.7446.

H1 non-conforming DHEXA



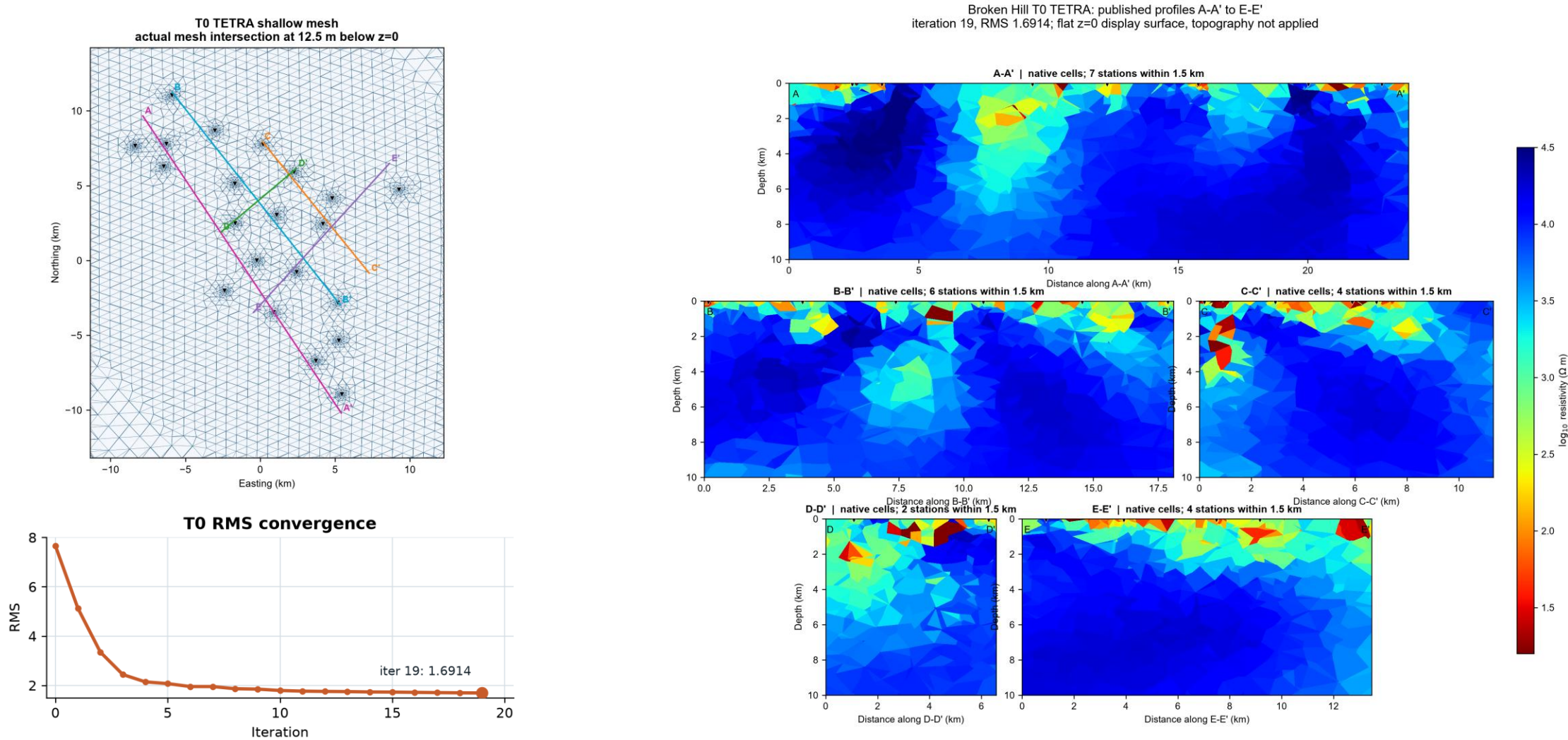
H1 uses 375 m refinement within 2 km and reached the controlled final state at iteration 11 with RMS 1.7771.

H2 nested non-conforming DHEXA



H2 uses 187.5 m refinement within 1 km and 375 m within 2 km, and converged at iteration 13 with RMS 1.7374.

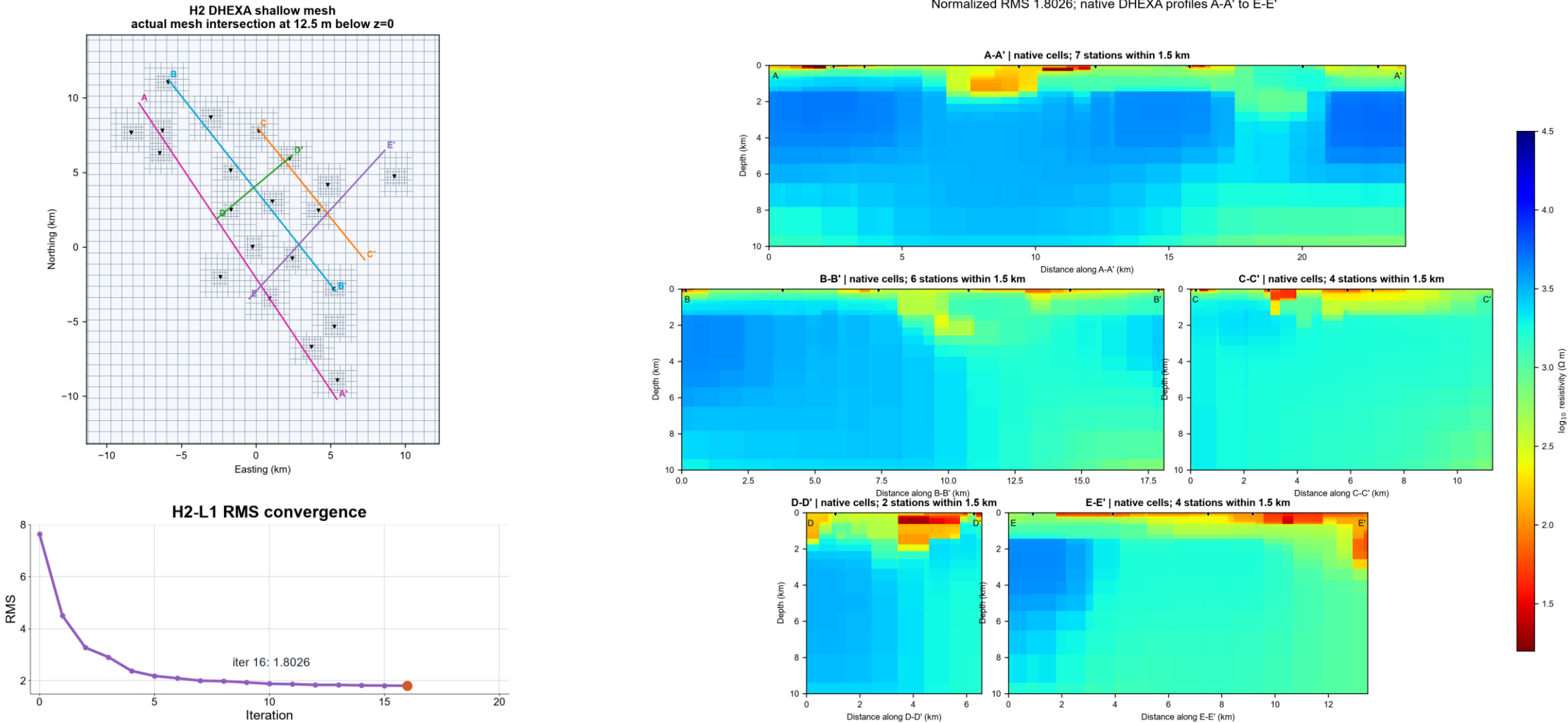
T0 tetrahedral mesh



T0 uses the final realized tetrahedral mesh and converged at iteration 19 with RMS 1.69145.

H2 Difference-L1 regularization

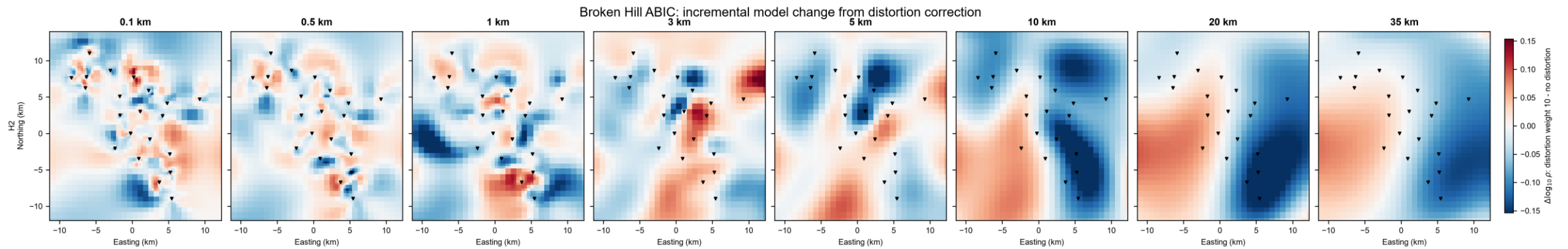
Mesh, norm, and distortion



Difference-L1 uses the H2 mesh and converged at iteration 16 with normalized RMS 1.8026; all five profiles are shown to 10 km depth.

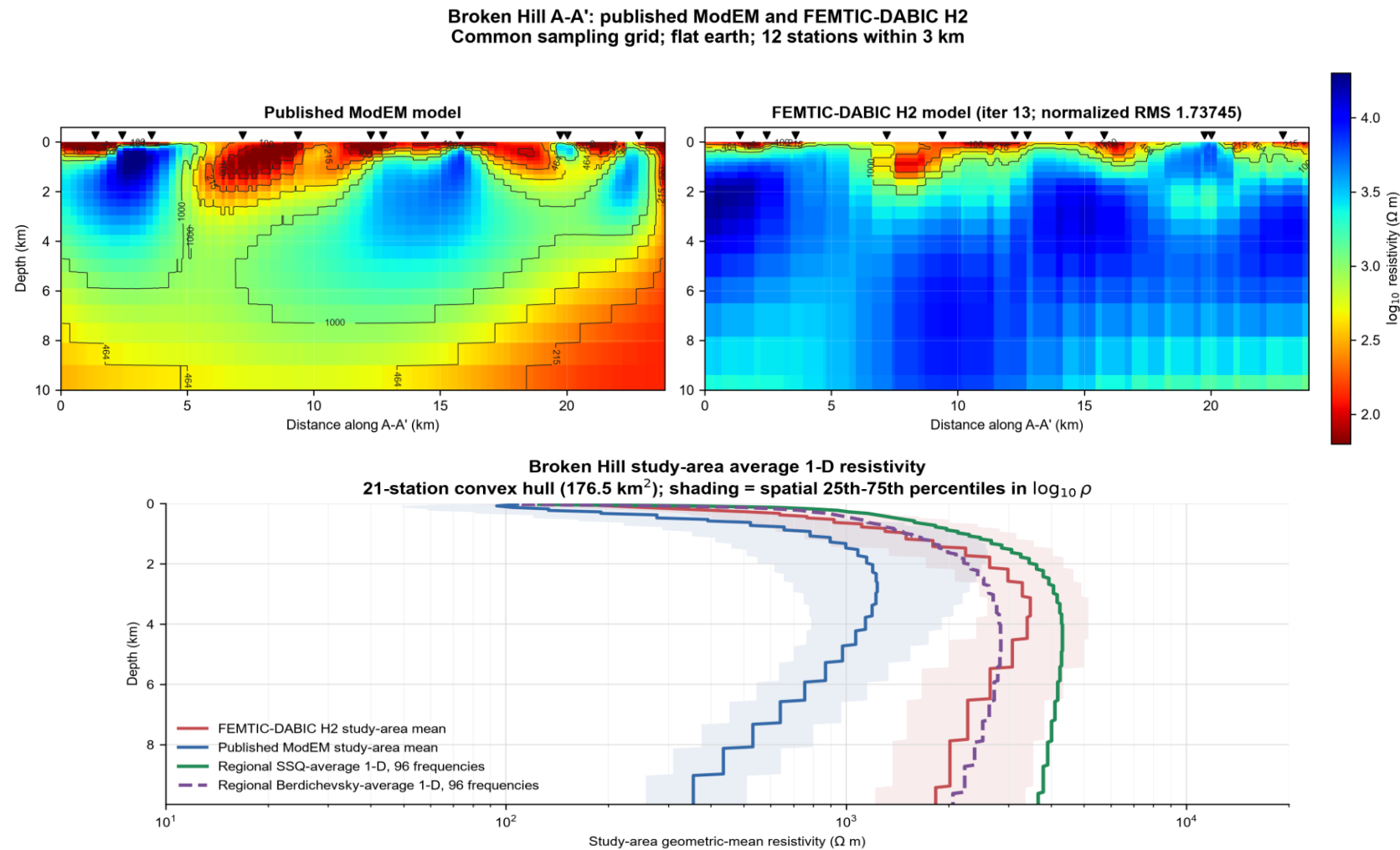
H2 galvanic-distortion sensitivity

Mesh and distortion



For H2, distortion correction changes RMS by -0.0114 (-0.66%) and has a small effect on the displayed resistivity structure.

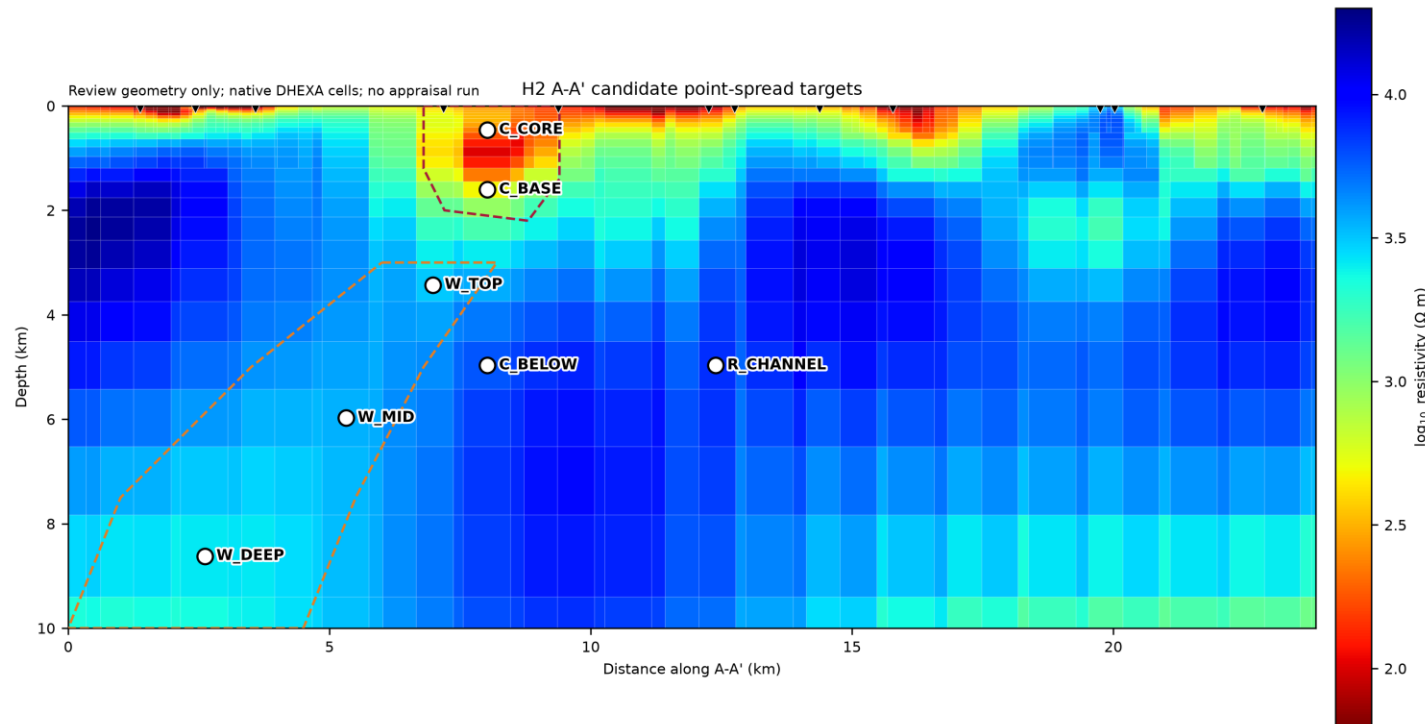
Published ModEM and H2 comparison



Published ModEM and H2 show partly overlapping conductive zones and similar broad depth trends, but differ in regional background, anomaly amplitude, and vertical extent.

Resolution analysis (point-spread function targets)

Model appraisal



Model-resolution matrix

$$R_m = H^{-1}G$$

$$H = J^T C_d^{-1} J + Q$$

$$G = J^T C_d^{-1} J$$

Point-spread function

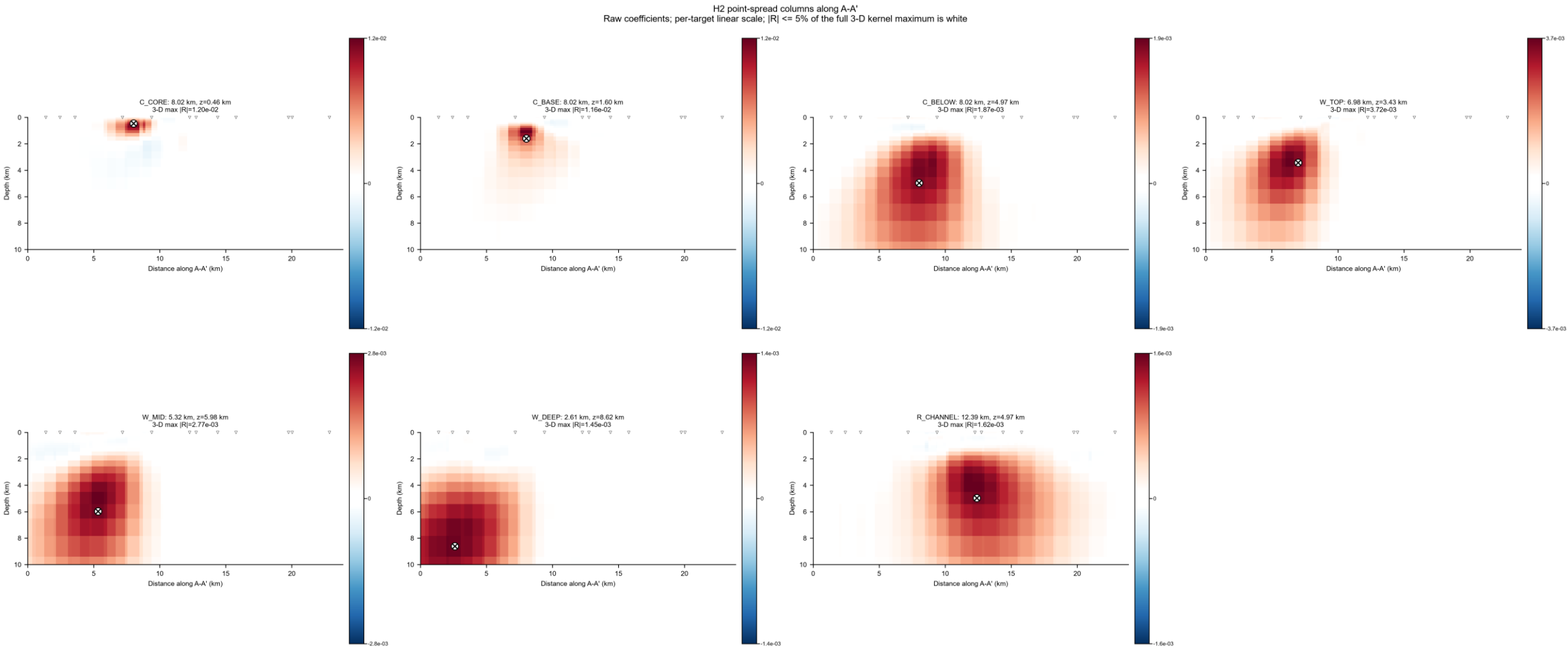
$$PSF_j = R_m e_j$$

PSF_j is the locally recovered image of a unit perturbation at model cell *j*. A compact kernel indicates better localization; a broad kernel indicates spatial mixing. This is a local linear diagnostic around the accepted H2 model.

The targets sample the shallow conductor, the inclined weak-conductive zone, deeper cells, and the right-side candidate channel.

Resolution analysis (point-spread functions)

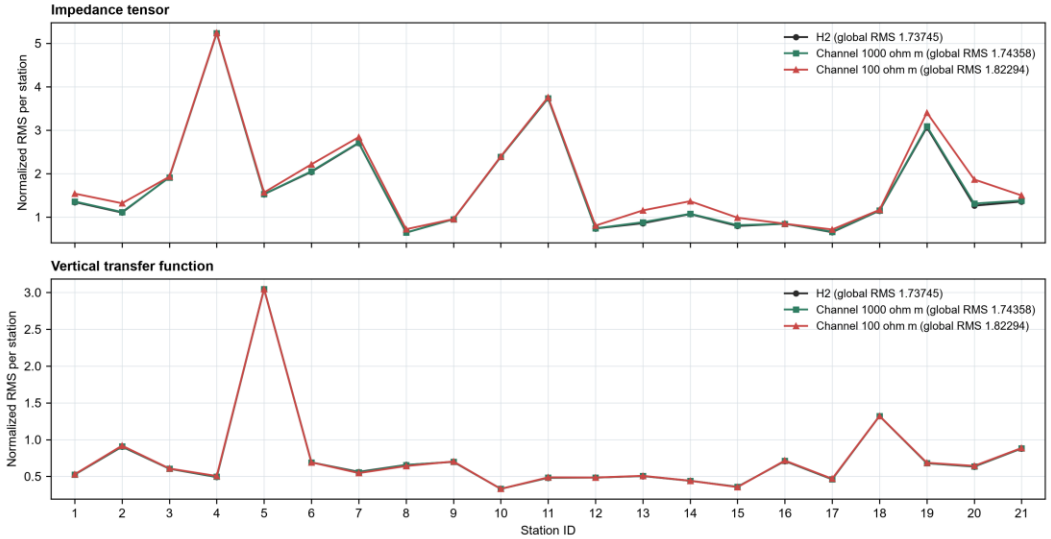
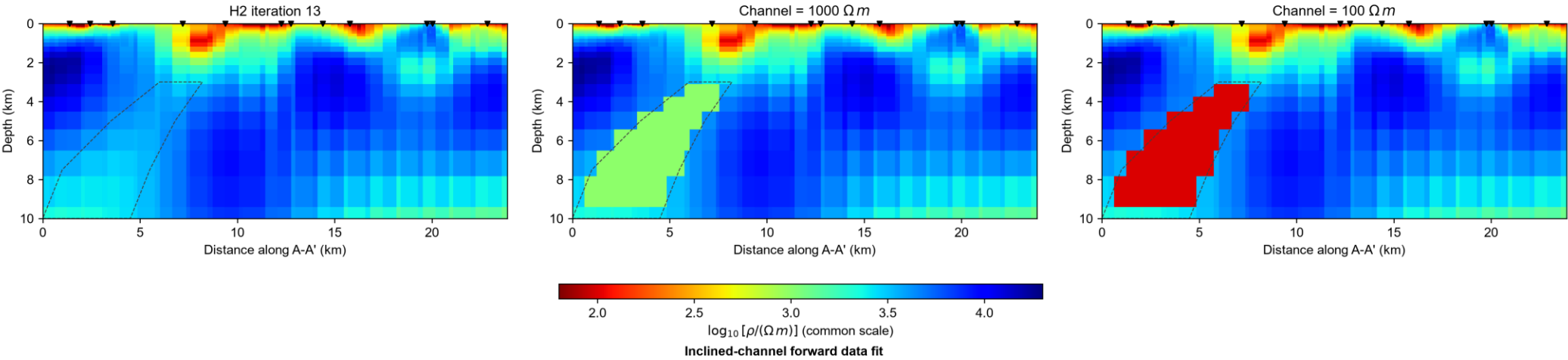
Model appraisal



Shallow targets are more localized. Deeper and off-profile targets are broader and weaker, so their boundaries and connectivity are spatially mixed.

Nonlinear forward test of the inclined channel

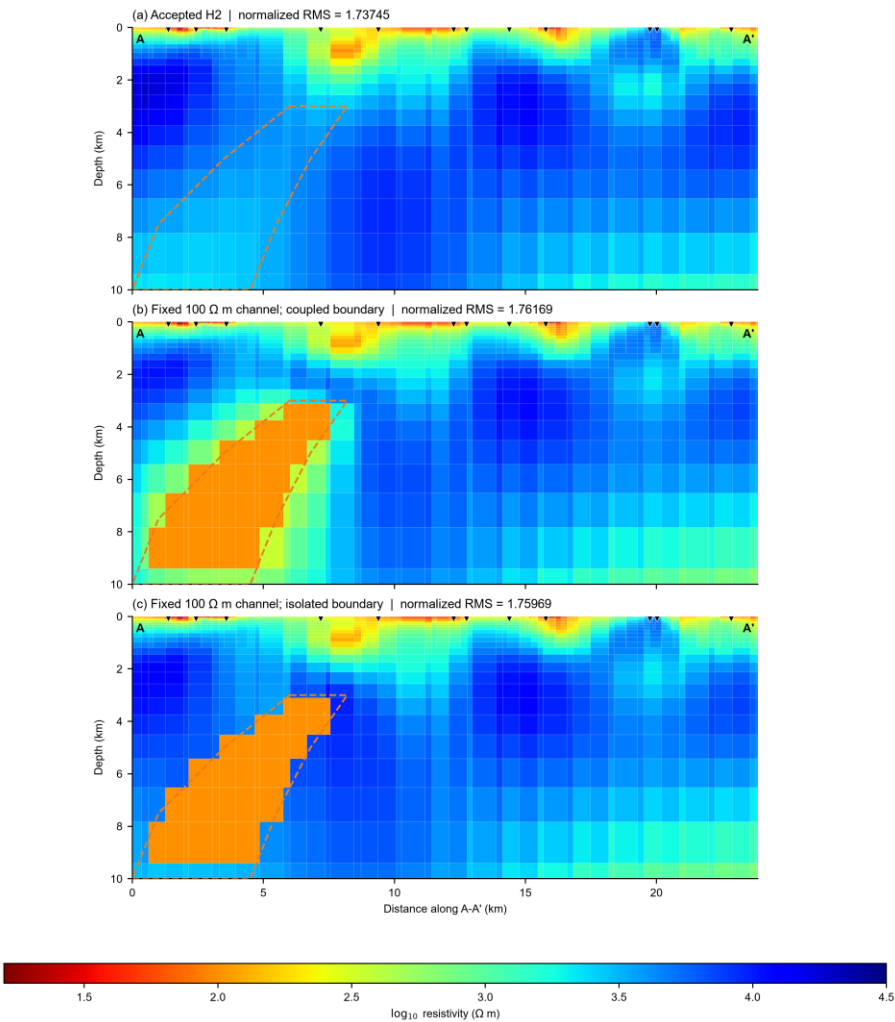
Model appraisal



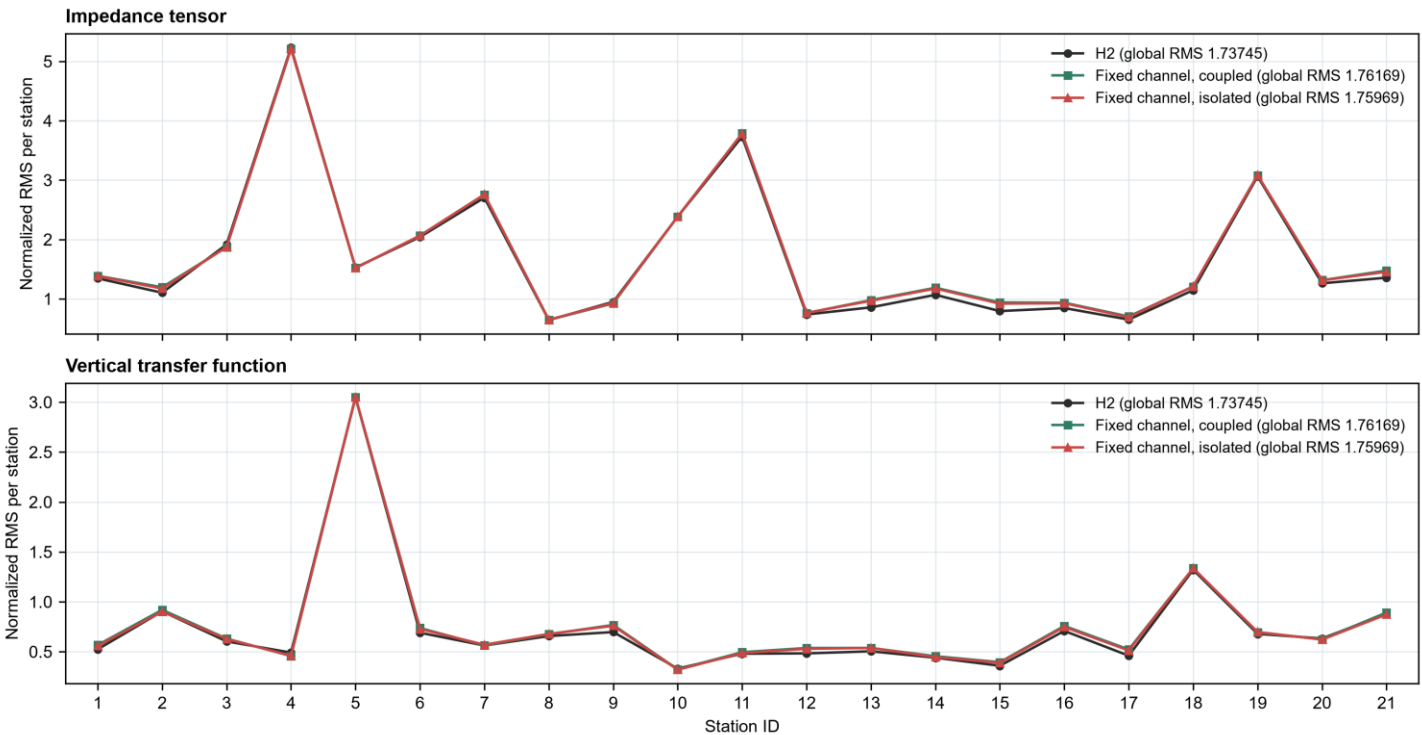
Without model compensation, global RMS increases from 1.73745 to 1.74358 for 1000 Ω m and to 1.82294 for 100 Ω m.

Constrained inversion of the inclined channel

Model appraisal



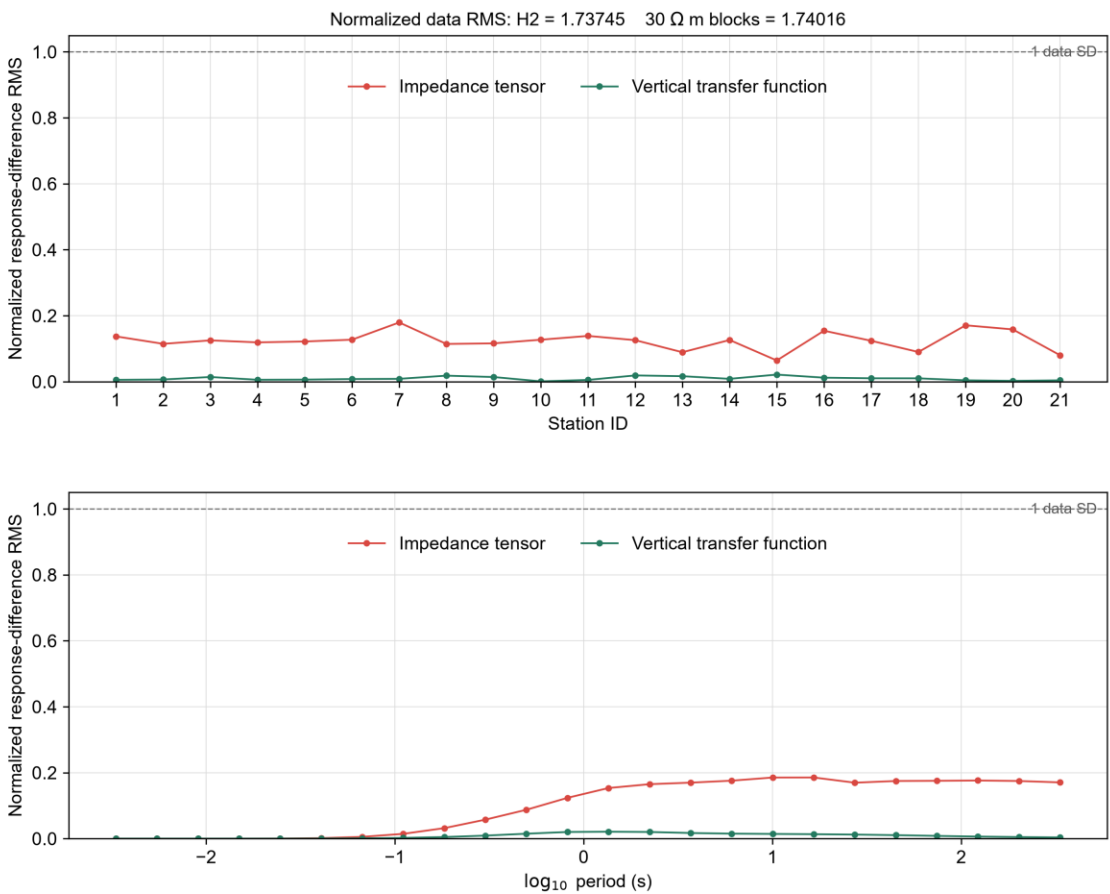
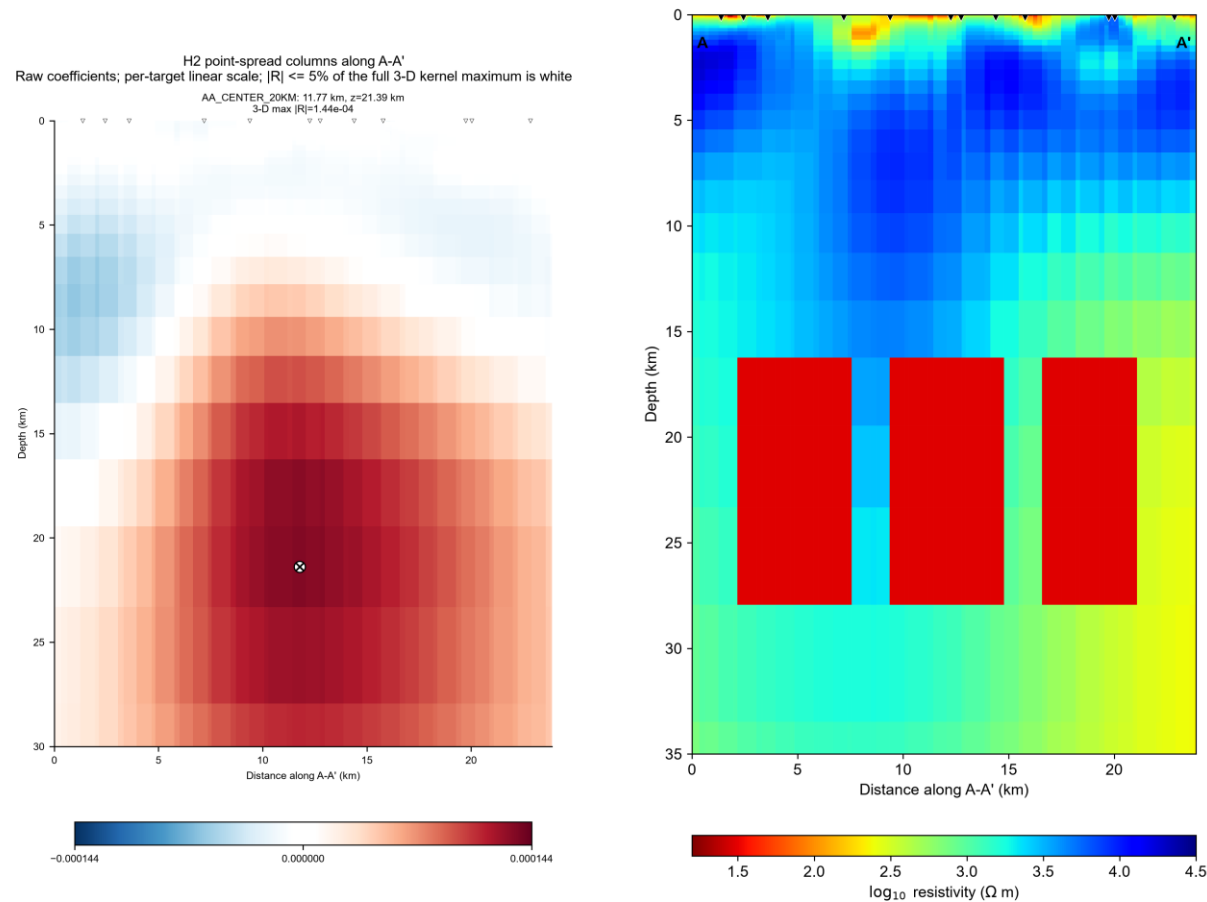
Constrained-inversion data fit



The prescribed 100 Ω m channel can be accommodated with only a ~1% RMS increase after re-inversion, with compensating changes elsewhere in the model.

Deep-conductor Resolution analysis and forward detectability test

Model appraisal



The deep PSF is broad. Three 30 Ω m blocks change normalized data RMS from 1.73745 to 1.74016 (+0.16%).

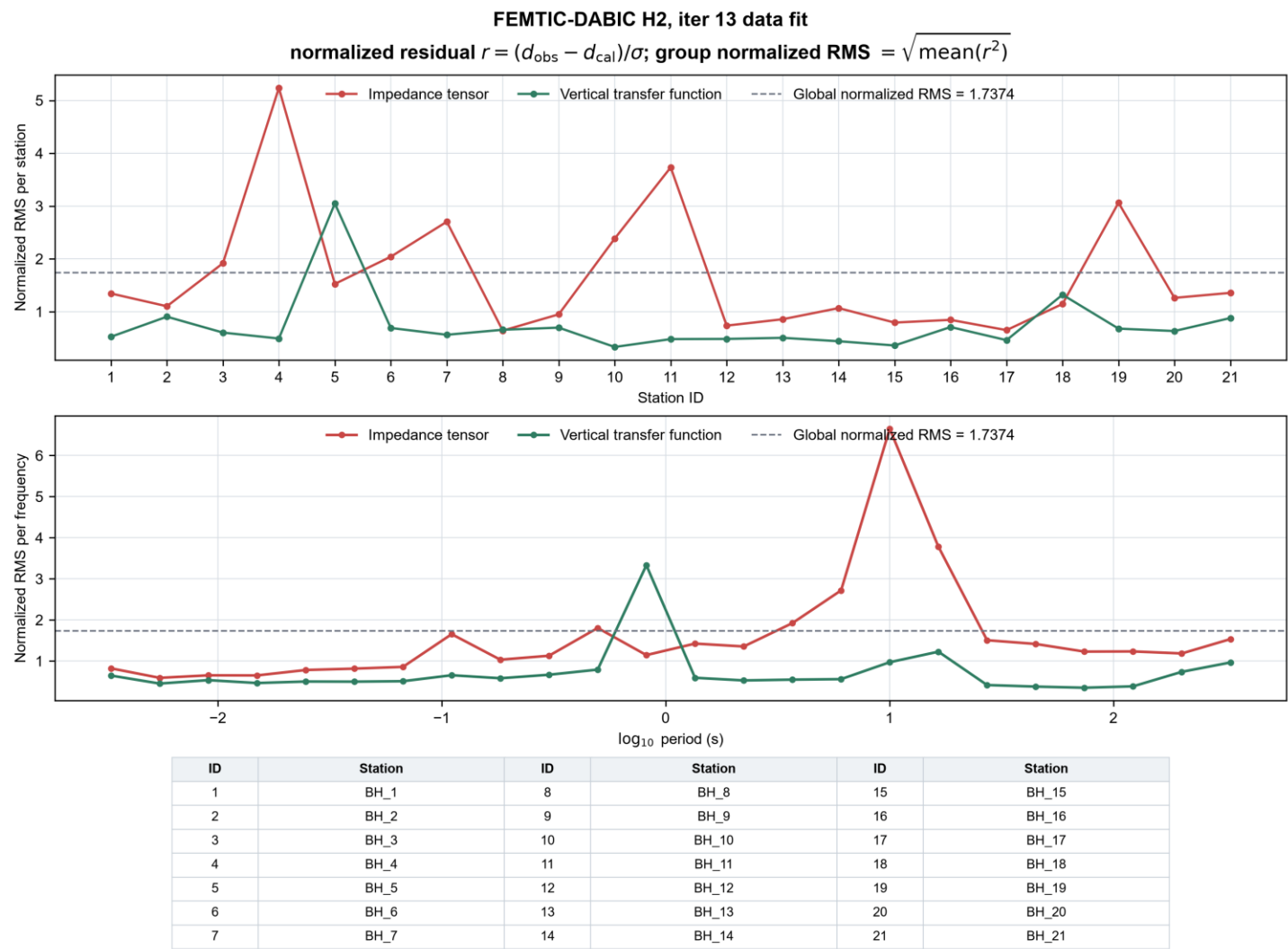
Conclusion

Limited data coverage and a low observation-to-model ratio make the inversion strongly underdetermined. Models therefore differ across inversion frameworks, meshes, regularization choices, and distortion settings.

The appraisal shows stronger localization for shallow targets and increasing spatial mixing at depth. The prescribed inclined channel near the northwestern end of A–A' (~0–8 km from A, 3–10 km depth) can be accommodated with only a ~1% RMS increase after re-inversion, with compensating changes elsewhere in the model.

The reduced ability to localize deep structures beneath A–A', illustrated by the ~20 km PSF and the prescribed conductors at 15–30 km depth, may partly account for the larger differences between Naser's and Sinan's deep models, as well as among the mesh tests in this study. Interpreting the deep structure in the context of a broader regional model, as done in Sinan's paper, may therefore provide a more robust basis for geological interpretation.

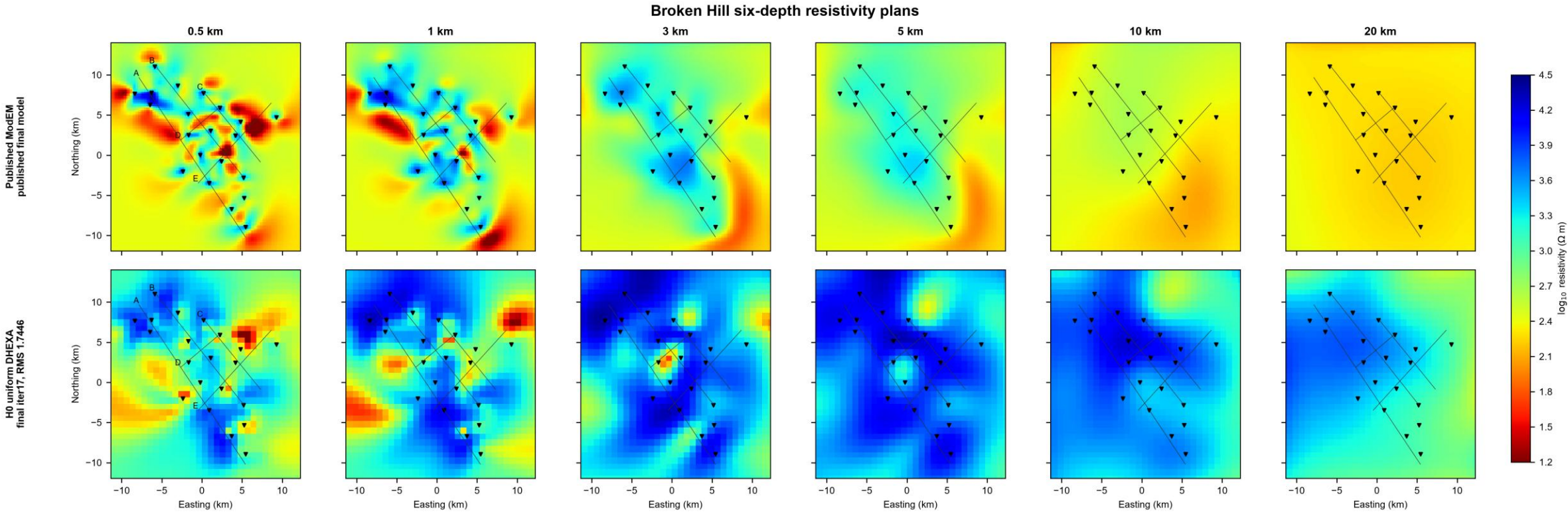
Appendix: H2 data fit by station and frequency



The accepted H2 global normalized RMS is 1.73745, while high values at several stations and periods warrant further inspection.

Appendix: Six-depth plans (Published ModEM and H0)

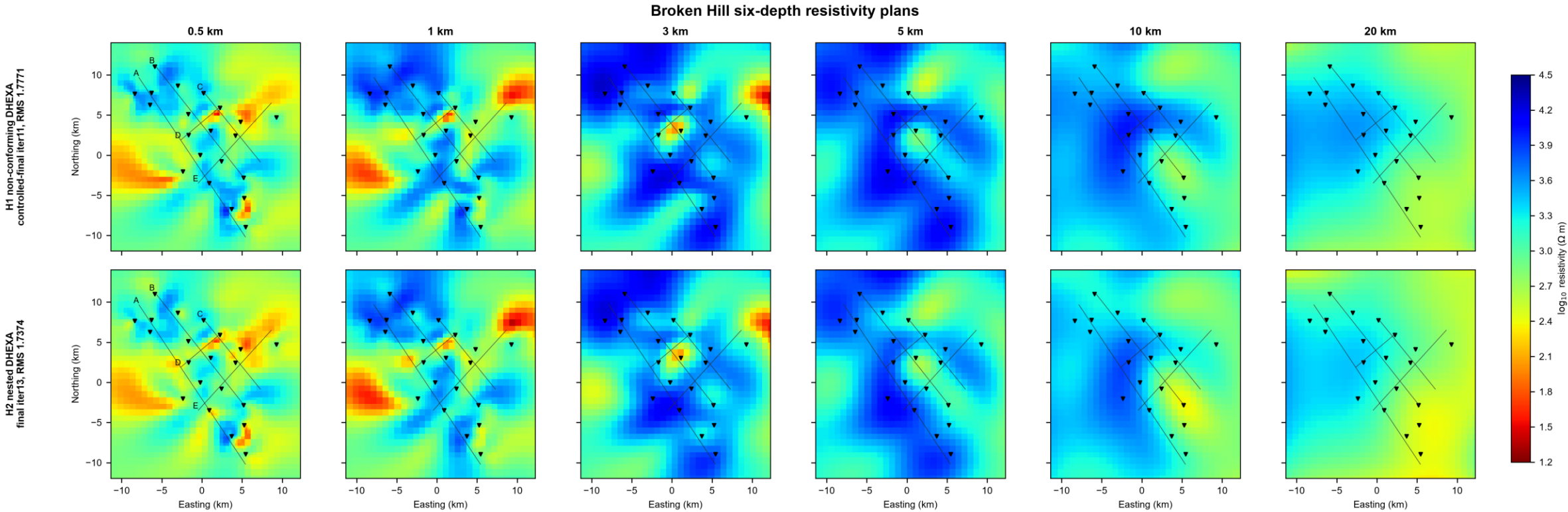
Appendix



Both rows use the same horizontal grid, profile traces, depths, and 1.2-4.5 log10 resistivity range.

Appendix: Six-depth plans (H1 and H2)

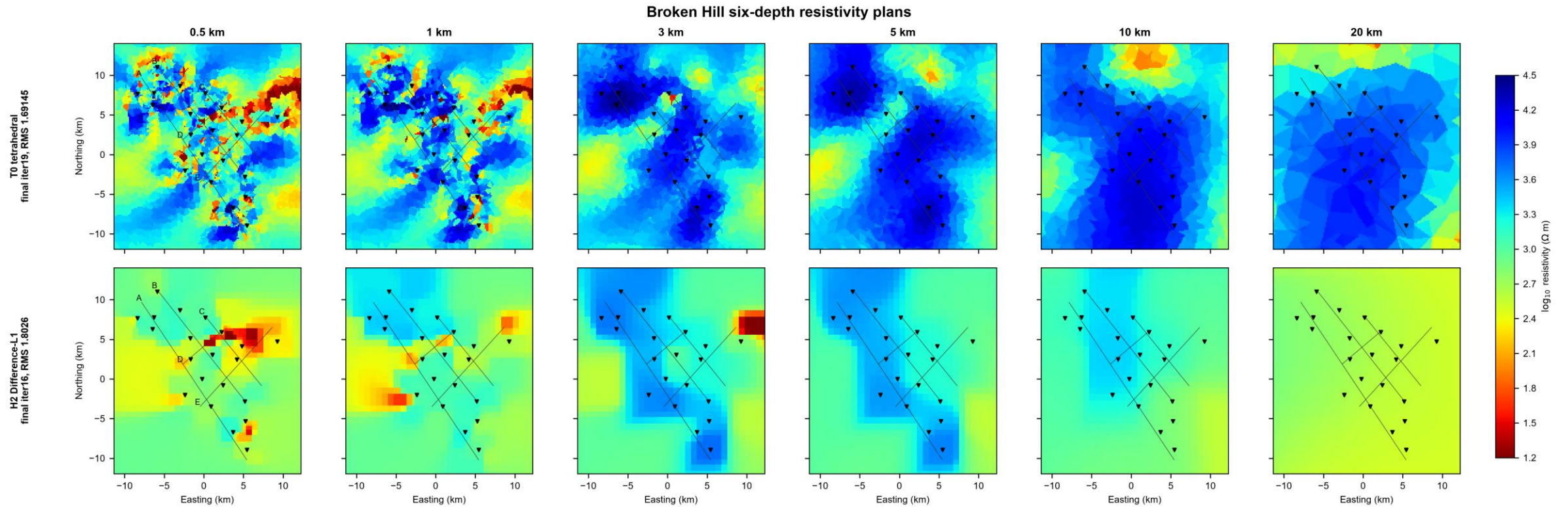
Appendix



H1 and H2 differ only in DHEXA refinement; both rows use the same sampling grid and resistivity scale.

Appendix: Six-depth plans (T0 and Difference-L1)

Appendix



T0 tests tetrahedral discretization; Difference-L1 tests the regularization norm on the H2 DHEXA mesh.